

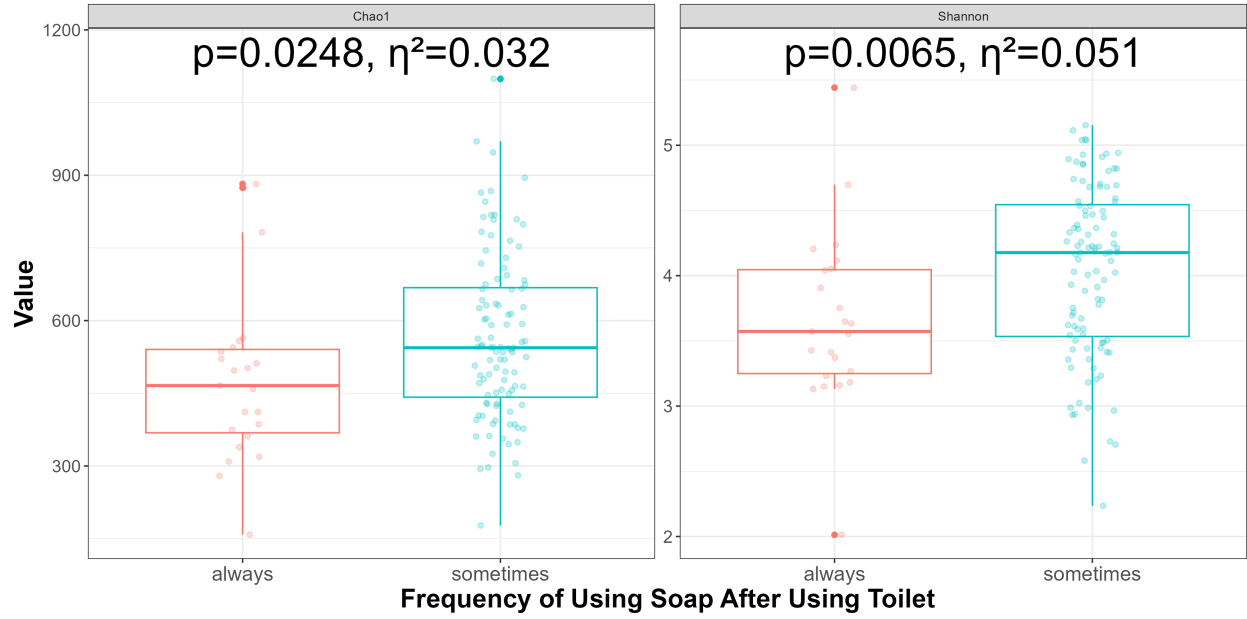


Figure 1: We calculated two metrics of alpha diversity for each child: Shannon and Chao1. The Shannon metric quantifies abundance and evenness of species in an ecosystem as a measure of diversity. Chao1 is an estimate of total species using singletons to infer unobserved species. Hypothesis testing values were done with a Wilcoxon signed rank test for equality of medians, a non-parametric alternative to a two-sample t-test. Across both metrics, the group of children that used soap only sometimes after using a latrine (toilet) had statistically higher levels of alpha diversity for Chao1 diversity ($p = 0.0248$, $\eta^2 = 0.032$) and Shannon diversity ($p = 0.0065$, $\eta^2 = 0.051$)

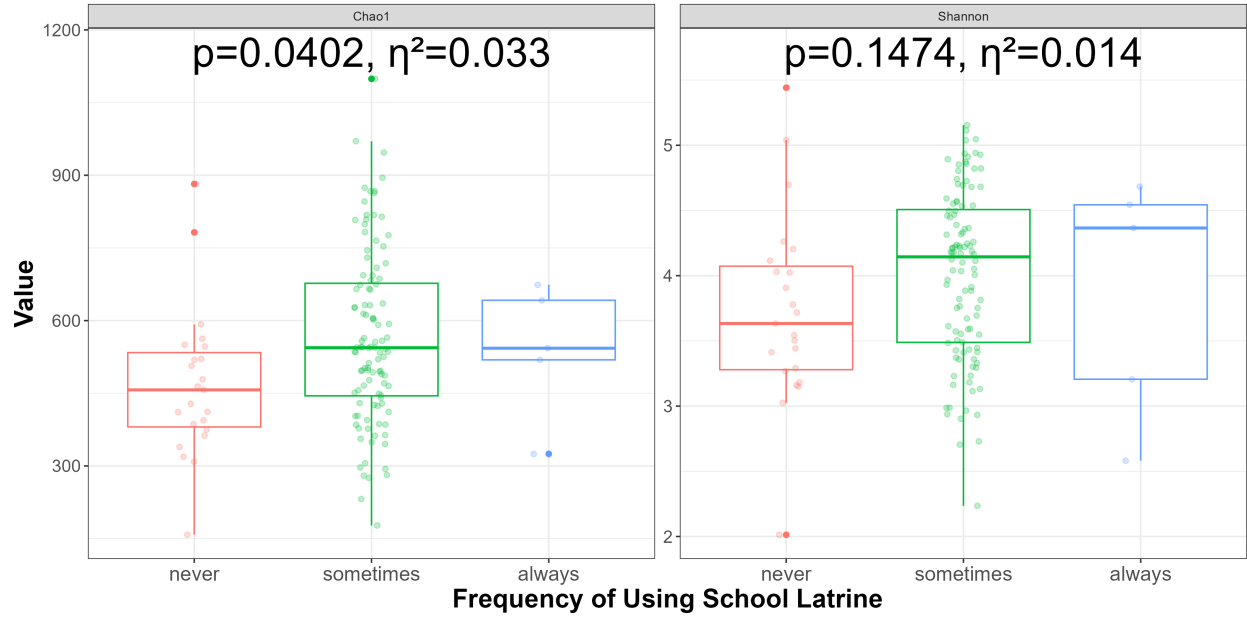


Figure 2: We calculated two metrics of alpha diversity for each child: Shannon and Chao1. The Shannon metric quantifies abundance and evenness of species in an ecosystem as a measure of diversity. Chao1 is an estimate of total species using singletons to infer unobserved species. Hypothesis testing values were done with a Wilcoxon signed rank test for equality of medians, a non-parametric alternative to a two-sample t-test. Chao1 diversity was significantly different across frequency of using a school latrine ($p = 0.0402$, $\eta^2 = 0.033$). Differences in Shannon diversity did not reach traditional levels of statistical significance.

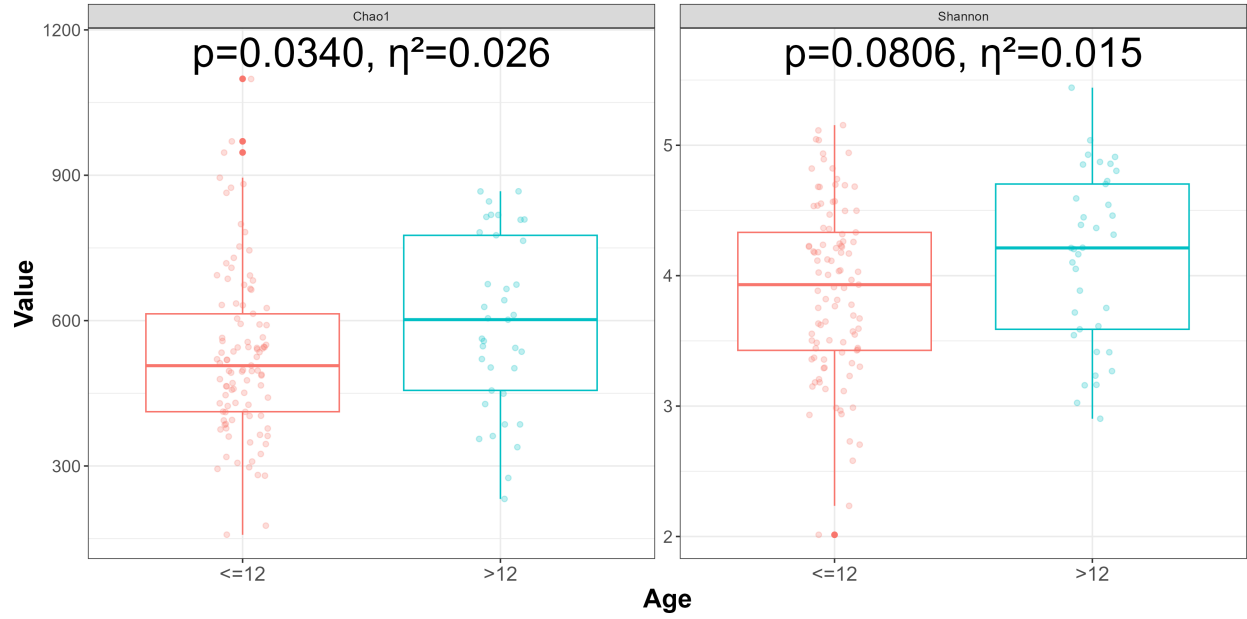


Figure 3: We calculated two metrics of alpha diversity for each child: Shannon and Chao1. The Shannon metric quantifies abundance and evenness of species in an ecosystem as a measure of diversity. Chao1 is an estimate of total species using singletons to infer unobserved species. Hypothesis testing values were done with a Wilcoxon signed rank test for equality of medians, a non-parametric alternative to a two-sample t-test. We found that children older than 12 had significantly higher Chao1 diversity ($p = 0.0340$, $\eta^2 = 0.026$) and marginally higher ($p = 0.0806$, $\eta^2 = 0.015$) Shannon diversity.

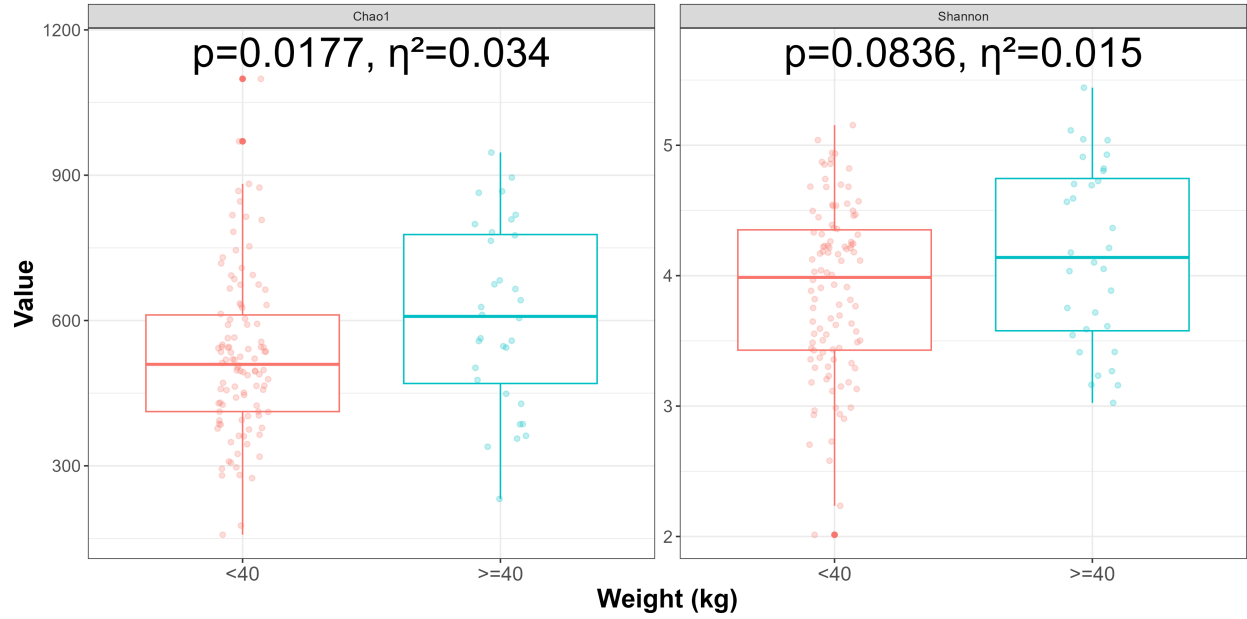


Figure 4: We calculated two metrics of alpha diversity for each child: Shannon and Chao1. The Shannon metric quantifies abundance and evenness of species in an ecosystem as a measure of diversity. Chao1 is an estimate of total species using singletons to infer unobserved species. Hypothesis testing values were done with a Wilcoxon signed rank test for equality of medians, a non-parametric alternative to a two-sample t-test. Children heavier than or equal to 40 kg had significantly greater Chao1 diversity ($p = 0.0177$, $\eta^2 = 0.034$) and marginally greater Shannon diversity ($p = 0.0836$, $\eta^2 = 0.015$).

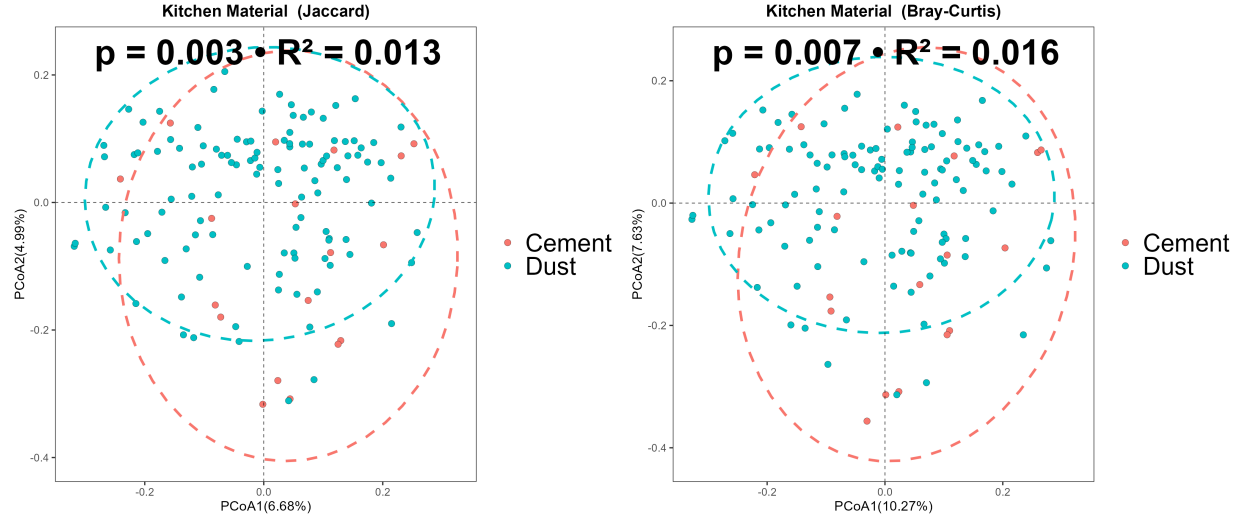


Figure 5: We calculated the beta diversity between groups of children who had kitchens made of dust/mud or cement. Beta diversity measures the difference in species composition between different groups and then plotted each group on a PCoA plot. A PCoA plot uses ordination to place an individual microbiome along two axes, with each axis representing a percentage of variation in diversity accounted for. The groups are then outlined with a 95% confidence interval ellipse. A PERMANOVA test of centroids and dispersion of the ellipses was used to assess differences in beta diversity between the two groups across two distance metrics (Jaccard & Bray-Curtis). Both distance metrics gave statistically significant beta diversity differences between the two groups (Jaccard: $p = 0.003$, $R^2 = 0.013$, Bray-Curtis: $p = 0.007$, $R^2 = 0.016$).

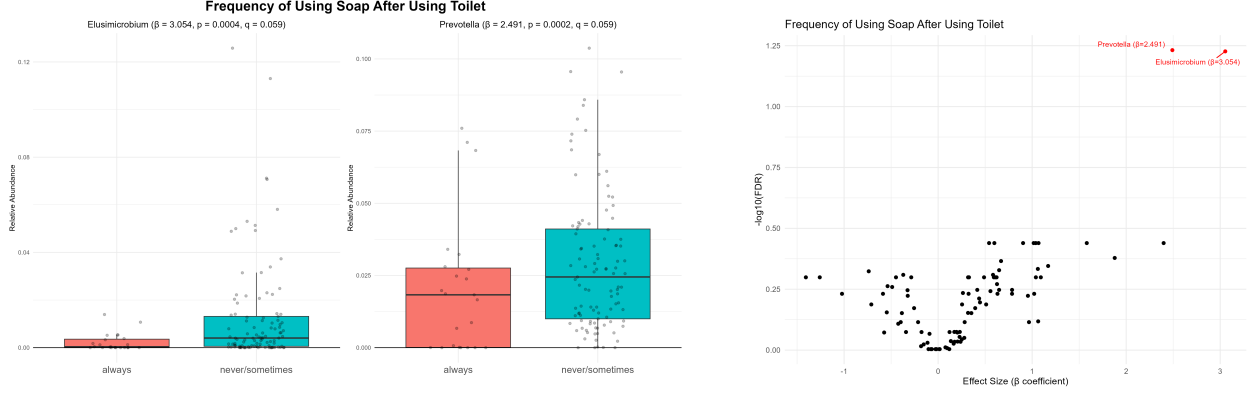


Figure 6: We performed Wilcoxon signed rank tests for equality of medians to test differential abundance of genera for children who always use soap after using a toilet versus children who sometimes/never use soap after using a toilet. Two genera were recognized as having the greatest differential abundance. We found that children who sometimes/never used soap after using a toilet had significantly greater abundance of *Elusimicrobium* ($\beta = 3.054$, $p = 0.0004$, $q = 0.059$) than children who always used soap. Also, children who sometimes/never used soap after using a toilet had significantly greater abundance of *Prevotella* ($\beta = 2.491$, $p = 0.0002$, $q = 0.059$). Both genera were only *marginally* significant after Benjamini-Hochberg p-value adjustment. On the right is the two genera highlighted on a volcano plot showing differential abundance effect size (x-axis) and p-value (y-axis, higher = lower p-value).

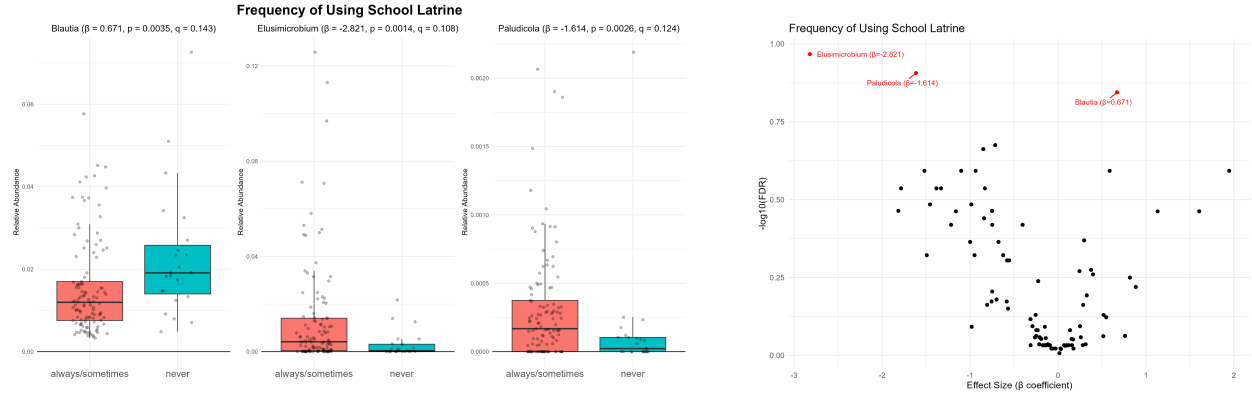


Figure 7: We performed a Wilcoxon signed rank test of equality of medians on the abundance of genera between children who never used the school latrine versus sometimes/always using the school latrine. Genera with Benjamini-Hochberg adjusted p -values < 0.15 were filtered down to 3 differentially abundant genera. Children who never used the school latrine compared to children who sometimes/always used the school latrine had higher abundance of *Blautia* ($\beta = 0.671, p = 0.003, q = 0.143$), lower abundance of *Elusimicrobium* ($\beta = -2.821, p = 0.001, q = 0.108$), and lower abundance of *Paludicola* ($\beta = -1.614, p = 0.003, q = 0.124$). None of the values remained statistically significant after adjustment. Also shown on the right is a volcano plot of each genera plotted according to their differential abundance effect size (x-axis), and their p -value (y-axis, higher = lower p -value)

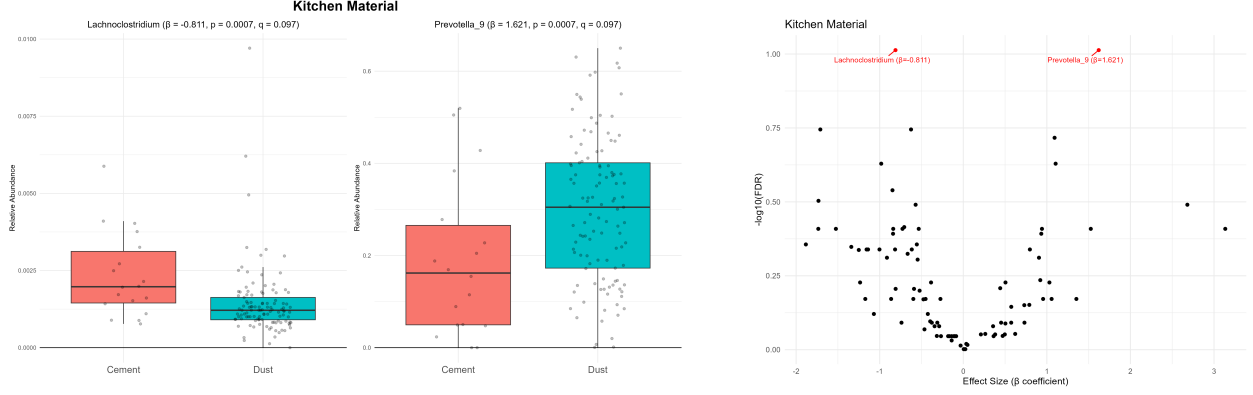


Figure 8: When comparing children with cement versus dust/mud floors in their kitchen, 2 differentially abundant genera were filtered with Benjamini-Hochberg adjusted p -values after performing a Wilcoxon signed rank test for equality of medians. Children with kitchen cement floors had significantly higher abundance of *Lachnoclostridium* ($\beta = -0.811$, $p = 0.001$, $q = 0.097$) than children with dust/mud kitchen floors. Conversely, children with dust/mud kitchen floors had significantly higher abundance of *Prevotella_9* ($\beta = 1.621$, $p = 0.001$, $q = 0.097$) than children with cement kitchen floors. Both genera were only *marginally* significant after adjustment. Also on the right is a volcano plot specifically noting the two differentially abundant genera in comparison to the effect sizes and p -values of the other genera.